Corporate Finance and Investment Science

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1 Financial Markets and NPV

Financial markets exist so companies and individuals can adjust consumption across time periods. The purpose is also to facilitate borrowing and lending. Investors can invest directly into fiems or via third parties called financial intermediaries which grant them flexibility.
financial intermediation can take three forms:

1. Size intermediation. Eg. banks combine investments of many small investors into large loans for a few borrowers.
2. Term intermediation. take from short term lenders (savings account) and re-lend to long term borrowers (mortgages).
3. Risk Intermediation. Intermediaries can assume the excess risk and protect borrower (hedge-funds investing into crypto)

Definition 1. *Project. Sequence of cashflows spread over time.*

To evaluate a project we can use several rules.

1.1 NPV

Theorem 1. ***NPV rule:*** *project is good if NPV ¿ 0. This implies the project is better than risk-free borrowing and lending in the market.*

Pros:

- Takes TVM into consideration, discounts all cashflows.
- Easy to evaluate NPV for combinations of projects.
- Can be applied to any weird pattern of cashflows.

Cons:

- Cashflows must be forecasted.
- What should the discount rate be?

- Doesn't take into consideration size of project.
- Doesn't take into consideration required projects, or projects with a differnt time horizon, like a choice between two required insurance plans.

Definition 2. ***Capital Budgeting*** *is the decision-making process for accepting or rejecting new projects/investments.*

Theorem 2.

$$NPV > 0 \iff \textit{increase in company value} \iff \textit{increase in shareholder value}$$

1.2 Payback Period Rule

Measures how long it takes to "payback" the initial investment of a project.
Pros:

- easy
- gives you a measure of liquidity.

Cons:

- Ignores TVM
- Ignores all cashflows after the payback period
- What is the comparison rule? is a payback period of 4 years good? ambiguity.

so in terms of comparing projects, this only does one thing right: give a measure of liquidity.

1.3 Discounted payback period rule

Number of periods in the future when the cummulative discounted cash flows total the initial investment. Pros:

- gives a measure of liquidity.
- Doesn't ignore TVM

Cons:

- Ignores all cashflows after the payback period
- Ambiguity of comparison rule

1.4 Internal Rate of Return

Definition 3. ***Cost of capital:*** *minimum RATE OF RETURN ("cost") needed to justify the "capital" needed to undertake a project.*

Definition 4. *Cost of capital such that the NPV of a project is 0*

Rule: Accept a project if IRR ¿ required rate of return. This "required rate of return"

1.5 Profitability Index

$$PI = \frac{\text{PV of future cashflows}}{Initialcost} = \frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0} = \frac{C_1 + C_2 + \dots}{|C_0|}$$

notes: A project has $PI > 1$ exactly when $NPV > 0$. Convince yourself: Well just look at it mathematically, $0 < NPV = C_0 + C_1 + \cdots \implies C_1 + C_2 + C_3 + \cdots > -C_0$ and thus the nominator of the Profitability index calculator is bigger than the denominator, thus $PI > 1$.
Interpretation: $PI = 1.2$ means that the future cashflows are 20% larger than the cost of the principal investment. So "loosely speaking," the project generates a 20% excess return over its life.
PROS:easy and correct for independent projects CONS: Scale. Maybe we don't care about percentage returns and only about absolute returns.

2 Capital Budgeting

What cashflows to include in analysis? *net incremental cashflows*, that is, cashflows that occur exclusiveley due to acceptance of the project (not the cost of toilet paper in the office). To identify a comprehensive list of the cashflows as yourslef:

What is the difference betweeen the cashflows of the entire firm *with* the project VS *without* the project?

We are interested in Cash flows, not accounting income.
We exclude:

- sunk costs: costs that ocurred in the past.
- depreciation
- goodwill: extra profit paid by other parties.
- unearned capital gains: appreciation of assets (taxable).

We include:

- Opportunity costs: cashflows that would have happened if the project were not accepted. ie if Rocky Balboa doesn't fight and decides to wash cars instead, he will get a carwashing salary but at the same time give up the opportunity and money that comes from fighting.

2.0.1 formulas

Operating Cash Flows (OCF): inflows and outflows related to business activity. Earnings before interest and taxes (EBIT)

$$OCF = EBIT - Taxes + Depreciation$$

$$EBIT = Revenue - Cost - Depreciation$$

$$Taxes = EBIT \times CotporateTaxRate = EBIT \times T_C$$

The project cashflows include initial cost when purchased (material asssets), and salvage value when fixed assets are sold (inflow) (sale used trucks, machinery, patents, etc at the end.)
OCF is independent of depreciation (it ends up cancelling in formula) because, intuitiveley, depreciation as an accounting concept to allocate the initial cost of the product over time, and not a cash cost.
NOTE: Capital cost: initial costs. Cost of capital: rate of return.

2.1 Capital Cost Allowance

In Canada, depreciation is NOT tax deductible. Instead the CRA allows a Capital Cost (initial investment) Allowance which provides a "tax-shield" which reduces EBIT and thus taxes.
Every year, the undepreciated capital cost (UCC) is reduced by the CCA rate which depends on the class of the asset at hand (ie $10, 10*(1-r), 10*(1-r)^2$), for instance cars may have a CCA rate higher than warehouses. And from that UCC we take a tax rate to get the actual tax shield amount: amount to be deducted from EBIT.
PVCCATS : Present Value of Capital Cost Allowance Tax shield.

2.2 Equivalent Annual Cost

Used when NPV misses the point. EACs spread the NPV of different projects to each year in a "financially equivalent way"

2.3 Inflation

$$(1 + Nominal) = (1 + Real) \times (1 + Inflation)$$

The CCA tax shields must be calculated in nominal terms, since once the asset is purchased, the entire CCA table is determined REGARDLESS of future inflation.

2.4 Risk Analysis

Contrary to prior examples, in real life, future cashflows are uncertain. Methods to evaluate project with uncertain cashflows: Decision trees, MC, etc.

Definition 5. ***Contribution Margin:*** *Reveals how much 1 unit of product sales contributes to net income.* $(Unitrevenue - unit\ variable\ cost) * (1 - T_c)$

Example a Doughnut sells for 1.5, costs .7 to make, its contribution margin is .8 times corporate tax on profit.

3 Return and Risk Measures

The financial market is more than just risk-free borrowing and lending. We can invest on instruments such as cash, stocks, ETFs
In general: High risk expects high return. Low risk low return.
to measure return, we consider "hypothetical return of selling now" include dividends of the current period and selling price. $Return = \text{Div}_{t+1} + (P_{t+1} - P_t) = \text{Dividend yield} +$ **Capital gains yield**
Holding period returns are geometric means.
Risk measures are variance and stdev/volatility. Drawbacks of volatility as a risk measure: symmetric penalty for loses/profits and it is a risk measurement around the centre of the distribution (ie symmetric around mean). Other measures of risk: Downside Semi-Variance, VaR, Conditional VaR. Risk can depend on highesr moments: skewness, kurtosis, etc. If R is normal then all higher moments can be calculated from the first 2 (ie mean and var)
Stock returns are NOT normally distributed.

4 Capital Asset Pricing Model (CAPM)***

4.1 Portfolio of two risky assets

We have a randon variable equation to describe the portfolio RV: $\pi = t * A + (1-t) * B$. A portfolio is a linear combination of risky (Random variable) assets. We wish to compute portfolio statistics, which is easy.
An important question is: What is the best weight distribution for the portfolio? ie what percentage should be invested on asset A and on asset B. To answer that notice that we can do a couple of things: we can maximize return and we can minimize risk as a function of weights t.

Definition 6. *The feasibility/opportunity set is the the set of portfolios that can be constructed from a given set of assets. This is enforced by the weights constraint* $\boldsymbol{w}^t \boldsymbol{e} = 1$. *Visually, it will be the inside of the hyperbola.*

Theorem 3. *The efficient set will belong to the frontier/border along the hyperbola. Any portfolio along the efficient frontier is "good enough," and the choice between efficient portfolios will depend on an individuals "risk tolerance."*

4.2 Efficient Frontier

turn attention to multi-asset portfolios. Most variables are intuitive: $\boldsymbol{\mu}, R, w, e$. Now, for Σ, recall that entry ij of the Variance-Covariance Matrix corresponds to $COV(R_i, R_j)$ and that diagonal elements correspond to variances.
Portfolio return:

$$\mu_P = \boldsymbol{w}^T \boldsymbol{\mu}$$

Portfolio variance:

$$\sigma_P^2 = \boldsymbol{w}^T \Sigma \boldsymbol{w}$$

A key fact is that diversification reduces portfolio risk.

$$\sigma_P \leq \sum_{i=1}^{N} |w_i| \times \sigma_i$$

How do we construct optimal portfolios, uniqueley identified by $\boldsymbol{w}^*$ for a given set of N risky assets.
Assumptions:

- Single period model, ie no dividends no compounding, just evaluating returns of assets A and B one year from now.
- Assets are perfectly divisible, ie $\boldsymbol{w} \in \mathbb{R}^N$
- No transaction costs and no taxes (this is a good assumption to always mention in our types of questions).
- max return, min variance, or both.
- mean returns and covariances Σ are known.
- Σ is positive definite. This means all securities in the portfolio are risky ie stocks.
- Not all expected returns are equal.

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Theorem 4. ***(Two-Fund Theorem)****: The formula $w_{opt} = w_m + \tau w_z$ states that any efficient portfolio can be replicated by two portfolios where w_m is the minimum variance portfolio, and w_z is self financing. These two portfolios can generate the whole efficient frontier.*

An investor seeking efficient porfolios needs only invest in combinations of two portfolios that are known to be efficient. So only two investment products provide a complete investment strategy.

4.3 Inclusion of a Risk-Free Asset

The assumption that Σ is positive definite means no security is risk-free. If there was a risk free-asset, Σ would have one row of all zeros.
Now let's set up the CAL introduction. First, consider the portfolio $P = (1 - w_0)*Q + w_0*r_f$, consisting of one risky asset Q and one risk-free asset with weight w_0 and return r_f. Recall that the risk free asset is not a random variable so the only quantity we have from it is its expected return ie the savings account interest. We have the following equations for the mean and variance of this portfolio.
Mean:

$$\mu_P = (1 - w_0) * \mu_Q + w_0 * r_f \implies \mu_P = \mu_Q + (r_f - \mu_Q) * w_0$$

Considering that only the risky asset Q contributes to variance we have volatility:

$$\sigma_P = (1 - w_0) * \sigma_Q \implies w_0 = 1 - \frac{\sigma_P}{\sigma_Q}$$

Combining these two equations by substituting w_0 from (2) into (1) we have the **Capital Allocation Line (CAL)**. Note that it requires a couple lines of algebra to show.

$$\mu_P = r_f + \frac{\mu_Q - r_f}{\sigma_Q} * \sigma_P$$

Now you may be thinking, how is this useful for computations since the portfolio is pretty much just an affine function of the risky asset? Well the CAL is not used for computations, but to examine its feasible solutions on (μ_P, σ_P) space. This is because the variables r_f, μ_Q, and σ_Q are presumably known and thus we are left with the equation of a line. That being said, the CAL's are a amily of lines indexed by the risky asset. IMPORTANT: On (μ_P, σ_P) -space we got the efficient frontier which is a horizontal hyperbola. Inside of it lie all feasible portfolios, so for any point inside this convex set we have a unique CAL. We have not yet found an optimal portfolio until we talk about CML.
The slope of the CAL is called the **sharpe ratio:**

$$\frac{\mu_Q - r_f}{\sigma_Q}$$

What does the sharpe ratio represent? Recall that the numerator is the difference in returns of the assets in the specific portfolio: the risky return minus the risk-free return, and so it represents the "excess return." The Sharpe ratio in turn represents the **"excess return per unit risk"**. For risky asset Q the sharpe ratio it literary tells us how much better or worse its return is than a savings account for its risk.
For a given risk-free asset return (intercept), the steepest CAL is referred to as the Capital Market Line (CML) and it is the one with intecept r_f that is tangent to the efficient frontier

Where the capital market line is tangent to to the efficient frontier, we got ourselves the statistics of what is called the "tangent portfolio."

Theorem 5. ***One-Fund Theorem****: There is a unique optimal risky portfolio M, charachterized by $w_1, w_2, \ldots, w_N$ such that any efficient portfolio can be constructed as a combination of M AND the risk-free asset r_f by varying w_0. This creates a NEW efficient frontier which forms a line called the CML.*

Note that the tangent portfolio changes with r_f. If r_f increases, then σ_M and μ_M increases and the slope of the CML decreases.
So there are two stages and two *subsequent* efficient sets to portfolio selection. This is called the **"SEPARATION PRINCIPLE"** since the determination of the optimal risky portfolio , M, is separated from teh preferences of the individual.

- Determine the tangent portfolio M. ie determine how much of each risky stock is optimal.
- Decide the optimal mix of tangent portfolio and the risk-free asset based on investor's risk-preferences.

4.4 Capital Asset pricing Model

The diversification effect says that by adding securities toa portfolio we can reduce its overall risk. In general, risk cannot be reduced to zero.
Two sources of risk of a security:

- Systematic risk: market risk, non-diversifiable risk. The economy/market/system itself might be at a low or go belly up.
- Non-systematic/idiosyncratic/diversifiable risk. Invest in Oil and Tesla, S&P 500 and Gold.

We can only diversify away non-systematic/idiosyncratic risk. A key consequence is that **investors should only expect to be compensated for systematic risk** ie the risk of cars, the risk of crypto. Investors **should not expect tor receive excess return for assuming non-systematic risk** since that can be diversified away. So if i'm a hedge fund manager, investors expect me to bet on all outcomes. If I have a crypto portfolio, they are rewarded for the risk of the crypto market, not for the risk of owning individual coins.... **NOTE*** feels like there's more to this.
So, given a security A, it has a certain, inherent amount of ideosyncratic risk: the cars and oil market determines how BMW does and that can't be changed. The systematic risk of a security is the market risk of that security, ie how it behaves compared to the market as a whole.
So, to quantify the systematic risk of security A, we need the know its correlation with the market portfolio.

Definition 7. ***Market portfolio @TODO DEFINE !!!!!???/***

Given a security A and a market porfolio M, we define the securitiy's beta, β_A by:

$$\beta_A = \frac{Cov[R_A, R_M]}{Var[R_M]}$$

A security's beta is a *measure of the systematic risk of the security.*
The CAPM formula:

$$\mu_A = r_f + \beta_A * (\mu_M - r_f)$$

Intuitiveley this quantifies how the return of our favourite asset can be determined from the market return.
Note the risk premiums:

- Market risk premium: $\mu_M - r_f$
- Risk premium of Asset A: $\beta_A * (\mu_M - r_f)$

CAPM Assumptions:

- All investors are rational, mean-variance optimizers.
- Investors plan for a single horizon.
- All assets are **publicaly** traded.
- All information is publically avaliable.
- No taxes or transaction costs.

Note that CAPM tell us the expected return of an asset independent of its volatility. This means that, **in a diversified portfolio!!**, the toal risk of an asset σ_A is irrelevant. Since we are able to diversify all the non-systematic/ideosyncratic risk away, only the systematic risk, captured by β_A, matters.

4.5 Security Market Line (SML)

The CAPM generates a line in the $(\beta, \mu)-$plane called the Security Market Line (SML). The CAPM is uniquely charachterized by the market and the risk-free rate (intercept r_f, slope $\mu_M - r_f$). In theory, **all securities lie on the SML line**. By plotting actual securities and comparing them to their expected location on the SML, we can determine if a security is correctly priced.

- If a security is **above** the SML, it has an actual return greater than expected for the given systematic risk measure β so it is **underpriced**
- If a security is **bellow** the SML, it has an actual return lower than expected, so for the given systematic risk it is overpriced (underdelivering, we'd expect more for that risk).
- On the line = correctly priced.

Definition 8. *The Traynor ratio of a security is a performance metric*

$$\frac{\mu_A - r_f}{\beta_A}$$

It measures the excess return of a security, for each unit of systematic risk, as given by beta. All securities on the SML have the same Traynor ratio. **Securities with a higher Traynor ratio have excess risk-adjusted returns, and can be identified as being underpriced.**

4.5.1 finding beta in practice

We don't compute $cov(R_M, R_A)$, because first of all it would have to be estimated, and second it might be hairy. Instead, we extend the CAPM formula from:

$$\mu_A - r_f = \beta_A * (\mu_M - r_f)$$

to a estimator version where we regress on observed returns of both a market proxy, like the S&P 500, (R_{M_i}) and our favourite asset (R_{A_i}). We consider the regression equation:

$$R_{A_i} - r_f = \alpha_A + \beta_A(R_{M_i} - r_f) + \epsilon_{A_i}$$

Note that in this equation, we observe R_{A_i} and R_{M_i}. Our residuals are ϵ_{A_i}, and we are regression to find parameters α_A and β_A. In theory, $\alpha = 0$ since all securities should lie on the SML, but this is not always the case so we call it **Jensen's alpha**. If $\alpha_A > 0$ then our security has excess risk adjusted returns above that predicted by CAPM. This is why investors are always "seeking alpha."

5 Arbitrage Pricing Theory (APT)

It's all about expected returns.
The CAPM formula states that the expected return on a security is a function of the return of the market portfolio. Put another way, the returns on securities are a function of the performance of the market. The APT extends this idea to include **additional factors**.
Start with the obeservation that the actual return of a security can be decomposed into two parts.

$$\text{actual return} = \text{expected return} + \text{unexpecte return}$$

Unexpected return due to risk can be further decomposed into two parts: Risk = Systematic + Ideosyncratic. The idea is that there are unexpected events which affect all securites (systematic risks) like war or natural disasters, as well as unexpected events which only affect a specific security: ie the death of a farmer affects farm stock.

APT model:

$$R_A = \mu_A + m_A + \epsilon_A$$

Where:

- Actual return R_A
- Expected return μ_A
- Excess systematic return m_A
- ideosyncratic return ϵ_A

5.1 Factor Model

APT uses a factor model **for systematic risk**.
Assume there are k factors $F_1, F_2, \ldots, F_k$ influencing the excess systematic return of a security.
The factor betas $\beta_{A,i}$ specify the sensitivity of asset A to factor i.

$$m_A = \beta_{A,1}F_1 + \cdots + \beta_{A,2}F_2$$

So the expanded APT model:

$$R_A = \mu_A + (\beta_{A,1}F_1 + \cdots + \beta_{A,2}F_2) + \epsilon_A$$

Note that factors model assumes linear sesnsitivity to risk factors.
Some common factors:

- Macro economic factors: GDP, inflation, oil, gold.
- Natural disasters: earthquakes, floods, global warming.
- socio-political: fiscal policies, warfare, etc.

Common assumptions:

- $E(F_i) = 0$ and $E[\epsilon_A] = 0$ so that all the "expected return" is captured in the μ_A of the same name.
- Assume factors are independent. $Cov(F_i, F_j) = 0 \iff i \neq j$
- Assume ideosyncratic factors are independent to systematic factors, $Cov(F_i, \epsilon_A) = 0$.
- Assume ideosyncratic factors of different securities A, B are independent $Cov(\epsilon_A, \epsilon_B) = 0$. ie one farmer only produces one product, so if it dies it only affects that one product (oversimplified).

CAPM vs APT CAPM deals with expected returns, APT with actual returns, it looks at the unexpected returns that deviate from the predicted.

NOTE that throughout CAPM and APT we are looking at portfolio returns, which really are RATES of return. Unlike absolute returns, a rate is a valid measure to compare securities. This is why we call the "expected value" (really just a constant cuz the risk-free asset is not a Randon Variable, but OK!) of the risk free asset r_f and add it is added to $(1-t)E[R_A]$

5.2 Expected returns

Definition 9. ***The Market Model****: is a one-factor model where the marker is the only risk factor.*

$$R_A = \mu_A + \beta_{A,M}(R_M - \mu)M) + \epsilon_A$$

We can derive CAPM from the APT model. **@TODO pg. 50**
The connection is established in the following APT formula:
Assume an universe of N assets. Then the expected return on asset i satisfies.

$$\mu = r_f + \beta_{i,1}\gamma_1 + \cdots + \beta_{i,N}\gamma_N$$

the values $\gamma_i, i \in [1, \ldots, k]$" are the same for all N assets. These are called the "factor risk premiums."
NOTE: $\beta_{i,1}$ measures how much factor #1 affects asset i while γ_1 is how much factor #1 affects all assets equally.
Turns out that ϵ_A is a random variable with zero mean and some non-zero variance.
Finally if we use the law of large numbers (expected value of average is expected value of individuals as sum goes to infty), we get that in large portfolios, the idiosyncratic risk is all diversified away so ϵ_A

6 Risk, Return, and Capital Budgeting

The discount rate for a project is driven primarily by the **project beta** and not by the company's beta. So the interest of making a samsung phone will depend on the sensitivity of phones to the market and not so much on the sesnsitivity of samsung to the market... thinking of forecasting concepts, samsung and samsung phones are linearly dependent. Samsung phones has the most information, don't include unecessary info or it will emss up the regression... huh?

How do we compute the beta of a project? depends on the project obviously. If the project "looks like" an extension of the company, use comapny's beta. If it looks very different, use the industry beta.
If tim hortons has a project consisting of opening a new location... well that project pretty much is the embodiment of the company so use that beta. But if tim hortons is raising money for cancer, then use the cancer donations "industry" beta.